

Cross-correlation between length and position in real fracture networks

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[1] By analyzing local geometrical properties of a dense multiscale fracture pattern, we characterized statistically the correlations between length and position of a fracture. Apart from some resolution effects, we show that the mean distance between a fracture center and its nearest neighbor is correlated to its length l such as $d \sim l^{0.3}$. Likewise, the average area around each fracture center within which no other fracture portion is lying has an ellipsoidal shape whose factor of anisotropy is correlated to l . The long axis, along fracture strike, is correlated to l such as $d \sim l^{0.25-0.3}$. The short axis remains invariant, about equal to the equivalent distance in a random fractal. At small scale (i.e., less than 1 m which is bed thickness) the shield area is isotropic. In addition to refining of geometrical model of fracture networks, such observations place some constraints on the stress interactions that prevail during fracture growth. **INDEX TERMS:** 8010 Structural Geology: Fractures and faults; 8020 Mechanics; 3250 Mathematical Geophysics: Fractals and multifractals. **Citation:** Darcel, C., O. Bour, and P. Davy, Cross-correlation between length and position in real fracture networks, *Geophys. Res. Lett.*, 30(12), 1650, doi:10.1029/2003GL017174, 2003.

1. Introduction

[2] Among the various geometrical parameters that may characterize a fracture, it is well known that its length is a critical parameter for fracture growth. For instance, in the case of an isolated planar crack with no cohesion between the crack-walls, the crack-tip stress and the displacement fields depend on the stress intensity factor that varies as the square root of the crack length [Scholz, 1990]. Nevertheless, only few studies have tried to characterize the role of fracture length in the development of a fracture within a complex population of cracks. Segall and Pollard [1983] showed from field investigations that for a sparse population of sub-parallel joints, the longer joints effectively restrain the shorter cracks from propagation so that longer fractures tend to create a crack shield that inhibit the propagation of other fractures. Analyzing in detail an outcrop, Ackermann and Schlische [1997] showed also that smaller sub-parallel normal faults were anticlustered around larger structures. They related this spatial distribution to a stress-reduction shadow around the larger faults. These works suggest therefore that the distance between a given fracture and its nearest neighbors should depend on the length of the fracture. These results may be of particular interest to reconstruct stochastic models of faults and

fracture patterns for input into fluid flow and transport simulators [de Dreuzy et al., 2001; Doughty and Karasaki, 2002], but also to place bounds on physical models for fault and fracture evolution [Cowie et al., 1995; Renshaw and Park, 1997]. However, taking into account the history and complexity of fracture growth processes, it is clear that such a relationship can be assessed only statistically in a dense fracture population with multiple orientations. Such a statistical correlation between the position and the length of a fracture has been indeed measured for the San-Andreas fault pattern but poorly quantified [Bour and Davy, 1999].

[3] Here, we analyze at different scales the geometry of a well documented fracture pattern to investigate the possible statistical relationships between the length of a fracture and the distance to its nearest neighbors. The fracture network used for the statistical analysis is the Hornelen fracture pattern that is representative of typical joint systems in massive rocks, systems that are wide-spread in nature [Odling et al., 1999]. The data set consists of seven embedded fracture maps (Figure 1) whose sizes go from 18×18 meters up to 720×720 meters. Each map represents a sub-sample of the whole Hornelen fracture pattern, with a lower bound controlled by the resolution scale of the mapping method and a higher bound given by the sample size [Odling, 1997]. Each map consists of at least 2000 fractures, perpendicular to the bedding plane, that are well individualized so that fracture locations and fracture lengths are consistently determined. This extensive database has been already used to test the relevance for the fracture length distribution of a power-law model that has been found to be valid over almost three order of magnitudes [Odling, 1997; Bour et al., 2002].

2. Center-to-Center Distance Between Fractures

[4] The most convenient and simple measure to characterize the possible relationships between the length of a fracture and the distance to its nearest neighbors, is to consider the average center-to-center distance, $d_c(l)$, between a fracture of length l and its nearest neighbor. Whatever the distribution of fracture densities, the absence of a statistical correlation between the position of a fracture and its length implies there should be no relationship between the distance to the nearest neighbor, d_c , and the fracture length l [Bour and Davy, 1999]. In other words, d_c should be the same whatever the fracture length classes if the probability to observe a fracture at a given point and in a given environment does not depend on its length. In the following, d_c is calculated for each fracture of the network and $d_c(l)$ is obtained through the averaging of d_c for the different fracture length classes. Note that the measurement of $d_c(l)$ is meaningful only in a statistical sense when dealing with stochastic fracture networks such as natural systems.

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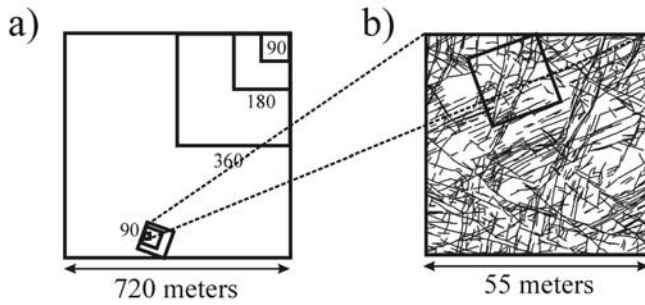


Figure 1. Sketch of the sampling procedure used to map the Hornelen fracture pattern [Odling, 1997]. (a) Relative location of the seven fracture maps. All the maps are included within the largest system of size 720×720 meters. (b) Examples of fracture trace maps ($T = 55$ m).

[5] In the present case, we do observe some variations of d_c depending on the fracture length l and on the system size T (Figure 2). The variations with the system size, T , that offset the different curves, are simply due to the limit of resolution that varies from map to map. Since the number of fractures is, at a first order, the same for the different fracture maps [Odling, 1997], the distances between fractures increase when one increases the size of the system. In addition, whatever the map considered, d_c increases also with l , implying a positive correlation between the fracture position and the fracture length. The larger the fracture, the greater is the distance to cover from the fracture center to its nearest neighbor. It implies that large fractures appear more isolated than small fractures that seem to be relatively more clustered.

[6] But, whatever the map, the positive correlation is observed only when $d_c(l, T)$ is larger than $\sim l/4$. For larger values of l , a saturation appears in all the curves with $d_c(l, T)$ about constant (Figure 2). In other words, the correlations between fractures appear to be hardly discernible when the distances from a fracture center to its nearest neighbor is smaller than a fraction of the fracture length. We believe that this effect is intrinsic to this point-to-point measure, which puts emphasis on the fracture center. If we assume that a correlation exists, and that it concerns a particular point on the fracture (for instance the initial nuclei), which is not a priori the fracture center, then a correlation is measurable only if the distance between fracture centers does represent the distance between nuclei. This occurs if the fracture is small enough compared to the distance between fractures. If not, the system should look like uncorrelated due to this uncertainty. This is indeed what is observed here where $d_c(l, T)$ is about constant for large values of l (for $d_c(l, T) < l/4$).

[7] The correlation measured on the different fracture maps for small values of l implies a relationship between $d_c(l)$ and l that is implicit in the case of space filling objects such as Apollonians [Mandelbrot, 1982] for which one expects $d_c(l) \sim l$. In opposite and as pointed out above, for a fracture system without any correlation between the fracture position and the fracture length, d_c should be constant whatever the fracture length. It should be equal in average to the mean minimal center to center distance between fractures, $d_{\min}$, that depends only on the system size T and on the number of fractures, $N(T)$, lying within that system [Bour and Davy, 1999]:

$$d_{\min}(T) = C \frac{T}{(N(T))^{1/D}}, \quad (1)$$

where D is the fractal dimension of the fracture network, and C a constant of proportionality that equals 0.5 for a two-dimensional non fractal random system ($D = 2$).

[8] Here, for each fracture map, it is possible to model the correlation observed for a limited fracture length range, by a power-law such as (Figure 2):

$$d_c(l) \propto l^\beta. \quad (2)$$

The exponents β obtained from each fracture map remain within the range 0.3 ± 0.1 whatever the map considered. The uncertainty depends on the range of lengths assumed for the fit but, whatever the range chosen, the exponents are clearly much smaller than the one expected for space-filling objects. It is also clear that for each fracture map, the distances do not vary significantly with l for the largest fracture length ranges.

[9] To check the consistency of our results with all fracture maps, we plot the normalized curve: $d_c(l, T)/d_{\min}(T)$ versus $l/l_{\min}(T)$ (Figure 3), where $d_{\min}(T)$ is the point-to-point distance in the absence of correlation (see equation 1), and $l_{\min}$ is the smallest fracture length range. Note that for the Hornelen fracture network, that corresponds to a self-similar system [Bour et al., 2002], $d_{\min}(T)$ is theoretically equal to $l_{\min}(T)$ so that we used the same normalization factor for both axes (Figure 3). For all the fracture maps, we obtain a unique global model that follows $d_c(l) \sim l^{0.3}$ at small lengths over approximately one order of magnitude. For larger values of $l/d_{\min}(T)$, $d_c(l)$ is approximately constant in agreement with the uncorrelated case.

[10] It is clear that an exponent of about 0.3 implies relatively weak correlations, but this is consistent with the expected range of values that should be between 0 (i.e., uncorrelated case) and 1 (the apollonian model). Note that we have also tested the relevance of the measured exponent by randomizing fracture lengths while keeping fracture positions fixed. This randomization gives an exponent equal to 0 as expected. Thus, despite the intrinsic difficulty to measure weak correlation, we believe that a value of 0.3 for the scaling exponent should be relevant. Note also that it is

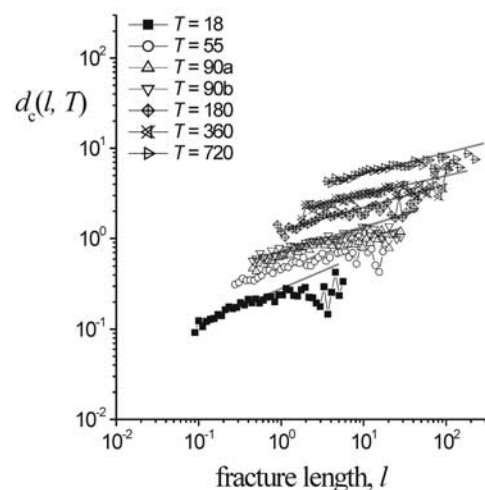


Figure 2. Computations of the average distance to the nearest neighbor, $d_c(l, T)$, for the seven fracture maps. The straight lines in gray correspond to power-law fits performed over the fracture length range such that $d_c(l, T) > l/4$. Scales are in meters.

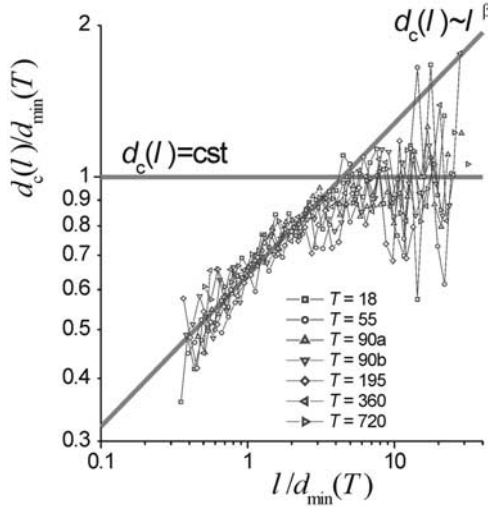


Figure 3. Normalized distances $d_c(l, T)/d_{\min}(T)$ versus normalized fracture length $l/d_{\min}(T)$ for the Hornelen fracture maps. The two gray lines that correspond to $d_c(l) \sim l^{0.3}$ and $d_c(l) \sim \text{cst}$, underline the two regimes observed for $d_c > l/4$ and for $d_c < l/4$.

consistent with the correlation measured on the San Andreas fault system for which we obtained a similar exponent [Bour and Davy, 1999].

3. Determination of the Anisotropic Scaling

[11] To go further in the characterization of the possible relationships between the length of a fracture and the distance to its nearest neighbors, we measured the average shape and size of the area surrounding a fracture of length l . This measure that characterizes the average local environment around fractures, has the great advantage to be less sensitive to the precise location of the nearest neighbor. In practice, we calculate the average distance between the center of a fracture and the first intersection with another fracture in the direction θ defined with respect to fracture azimuth (Figure 4a). This parameter, $d_f(\theta, l)$, that is measured all around fracture centers with $\theta \in [0, \pi]$, provides a direct estimation of the area around a fracture center into which no other fracture portion is lying. The distances $d_f(\theta, l)$ are calculated with a discrete step $\Delta\theta$ equal to 18° . The mean values of $d_f(\theta, l)$ are calculated for each θ over all fractures belonging to the same class of length. We arbitrarily fixed $\theta = 0$ in the direction of the fractures strike ($d_f(0, l) = d_{\parallel}(l)$) and $\theta = 90^\circ$ in the direction perpendicular to the fracture ($d_f(\pi, l) = d_{\perp}(l)$), as it is illustrated in Figure 4a. Not that for a uniformly distributed fracture network, without any statistical correlation between the fracture length and fracture position, we expect the average size of the non-fractured blocks, r_{block} , to be expressed through the relation:

$$r_{\text{block}} \propto \frac{T^2}{M(T)}, \quad (3)$$

where $M(T)$ is the cumulative length of the fractures included in the system of linear size T . In that case, d_f should not depend on the fracture length l , whatever the length of the fracture network. Note that the shape of the shield area may depend on the fracture orientation

distribution. However, whatever the orientation distribution, we do not expect any dependence of $d_{\parallel}$ or of $d_{\perp}$ if there is not any correlation between the length, the position and eventually the orientation of a fracture.

[12] The distance d_f between the center of a fracture and the first intersection with another one depends on the direction θ , the fracture length l and the system size T (Figures 4 and 5). Whatever map and fracture length, the distance $d_f(\theta, l, T)$ defines an ellipsoid whose long axis $d_{\parallel}(l, T)$ is parallel to fracture strike and whose short axis, noted $d_{\perp}(l, T)$, is perpendicular to strike (Figure 4b). The in-strike axis, $d_{\parallel}(l, T)$ increases with fracture length as a power law (Figure 5), $d_{\parallel}(l, T) \sim l^\gamma$, with $\gamma = 0.25 \pm 0.05$ for all maps except the 18x18m map for which the exponent is a bit smaller (Figure 5). In contrast the short axis $d_{\perp}(l, T)$ is about constant with l (Figures 4, 5) emphasizing an absence of correlation perpendicular to fracture strike. $d_{\perp}(l, T)$ is about equal to r_{block} defined in equation (3) (dotted lines Figure 5), in agreement with an expected behaviour for uncorrelated systems. Note that in terms of pure geometry, the smallest fractures mostly appear isolated and locally unconnected to others (Figure 4c) since their $d_{\parallel}$ and $d_{\perp}$ are larger than $l/2$. In opposite, we expect the largest ones to be locally connected to other fractures since $d_{\parallel}$ is in average smaller than $l/2$.

[13] Despite resolution effects, which can modify differently $d_{\parallel}$ and $d_{\perp}$, an isotropic point ($d_{\parallel} = d_{\perp}$) may be extrapolated from the trends measured on each map for $d_{\parallel}$

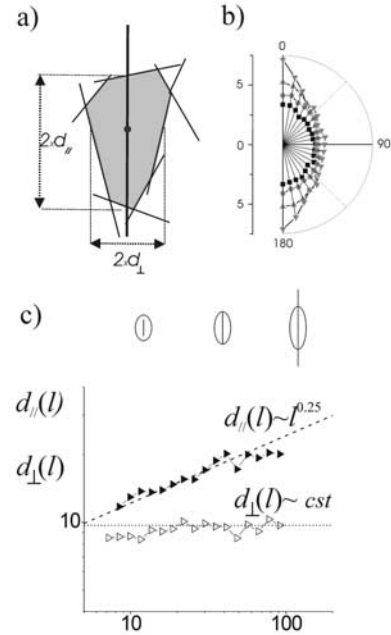


Figure 4. Characterization of the anisotropic scaling. (a) Schematic description of the area (in gray) surrounding a fracture into which no other fracture portion is lying. (b) Variations of $d_f(\theta, l, T)$ for several fracture lengths classes for the map $T = 180$ meters (squares: $l \in [1.8; 2.1]$, disks: $l \in [4; 4.7]$, up triangles: $l \in [8.8; 10.3]$, down triangles: $l \in [19; 22.8]$). (c) Variations of $d_{\parallel}(l)$ and $d_{\perp}(l)$ for the map $T = 720$ meters. Scales are in meters. Above the graph is drawn a schematic representation of the shield area shape with respect to the fracture length. For small fracture length $d_{\parallel} > l/2$, while for large fracture length $d_{\parallel} < l/2$, meaning that fractures are in average locally connected.

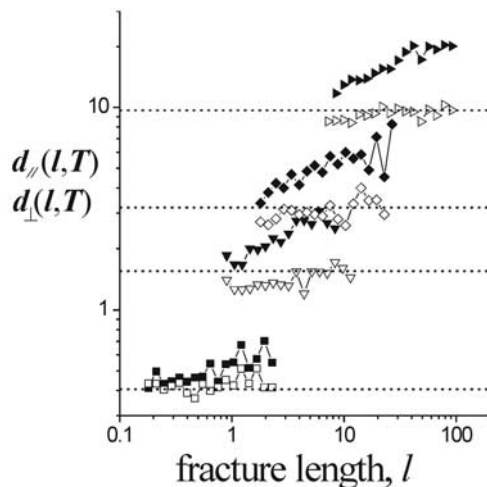


Figure 5. Variations of $d_{\perp}(l, T)$ and $d_{\parallel}(l, T)$ (see Figure 3a) for several fracture maps (full symbols, *resp.* empty symbols, represent $d_{\parallel}$ *resp.* $d_{\perp}$, and squares: $T = 18$ m, down triangles: $T = 90$ m, diamonds: $T = 180$ m, right triangles: $T = 720$ m). Scales are in meters. The results are presented only for four maps but similar results are obtained for all other fracture maps. Horizontal dotted lines correspond to the values of r_{block} calculated from Equation 3 for the four fracture maps.

and $d_{\perp}$. We found $d_{\parallel} \sim d_{\perp}$ for fracture length of about 1 meter whatever the map. This point is directly measured on the smallest map ($T = 18$ m) for which $d_{\parallel} = d_{\perp}$ when $l < 1$ m. This scale that is about bed thickness, corresponds also to the lower cut-off of the power-law fracture length distribution [Odling, 1997].

4. Discussion

[14] From a mechanical point of view, the distance to other fractures defines a kind of shield area that surrounds a fracture, whose shape is illustrated in Figure 4b. This shield area expresses both mechanical determinism due to stress interactions between fractures and stochasticity that may come from initial flaws for instance. The model of isolated fractures, with a stress enhancement at fracture tips and a stress decrease around fracture centre, would at least predict $d_{\parallel} \sim l/2$, and $d_{\perp}$ dependent on l [Segall and Pollard, 1983]. Clearly the results are inconsistent with such a simple scheme, and fracture interactions are likely to be dominated by much more complex interactions which produce long-range correlations responsible of fractal behaviour [Davy *et al.*, 1995]. The increase of $d_{\parallel}$ with l , and the fact that $d_{\perp}$ is constant with l and about equal to the average distance in random uncorrelated fractals, emphasize that most of stress correlation are along fracture strike.

5. Summary

[15] We investigated the local spatial correlations within the multi-scale fracture pattern of the Hornelen basin. We

show that the fracture length and the fracture position are two geometrical parameters that are statistically correlated for all the fracture maps. On average, the larger a fracture is, the greater is the distance to cover before encountering another fracture center. But our results suggest also that due to some intrinsic effects that should depend on fracture density, the characterization of such correlations remains nevertheless limited to a certain fracture length range. Whatever the maps and fracture lengths, the average areas around each fracture center into which no other fracture portion is lying have an ellipsoidal shape. The long axis is parallel to fracture strike, and is positively correlated to fracture length by a relationship of the form: $d_{\parallel} \sim l^{0.25}$. In contrast the short axis, that is perpendicular to fracture strike, is uncorrelated with fracture length and about equal to the average size of the non-fractured blocks. Such local correlations may be interpreted as remote consequences of the mechanical interactions between the fractures during fracture growth.

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